

Bhū-Darpan

AI-Powered Digital Twin of India's Climate
Jammu & Kashmir Pilot • Project Documentation

"Observe · Predict · Simulate — India's Climate, Mirrored"

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1. Executive Summary

Bhū-Darpan (“Mirror of the Earth”) is a working proof-of-concept **climate digital twin** of India, demonstrated over the Jammu & Kashmir pilot region, bundled with an AI satellite image-analysis module. It continuously mirrors the region’s climate state by fusing multiple national and open datasets, produces short-term AI forecasts of rainfall and temperature, lets users explore “what-if” scenarios, and detects climate hazards — all through an interactive web dashboard that runs offline on a single machine.

***In simple words:** It is like a living, digital copy of the sky and land over Kashmir. It shows what the weather/climate is doing now, predicts the next two weeks with AI, and lets you ask “what if it rains 30% less?” — then shows the impact on a map.*

Headline capabilities

- **Deep-learning forecasting** — a trained PyTorch LSTM predicts temperature at +15.7% skill vs climatology.
- **Multi-source data assimilation** — fuses NASA POWER (MERRA-2) and Open-Meteo (ERA5), with satellite land-surface temperature assimilated into the state.
- **Near-real-time** — one click pulls the latest available data and the map evolves.
- **Satellite AI** — an EuroSAT-trained CNN classifies land cover at 95.9% accuracy and detects change between two dates.
- **Interactive dashboard** — live state, geospatial maps, 14-day forecasts, what-if simulator, hazard intelligence.
- **Secure accounts** — email-OTP + password authentication persisted in SQLite.

2. The Problem

India’s climate information is powerful but fragmented. Ground observations come from the India Meteorological Department (IMD); satellite observations from ISRO platforms (INSAT via MOSDAC, Bhuvan, NICES); and global reanalysis from other agencies. These live in different formats, resolutions and portals, and conventional models are computationally heavy, coarse and not continuously updated.

- Data is siloed across IMD, INSAT/MOSDAC and reanalysis — hard to combine.
- Conventional climate models are slow, coarse, and rarely near-real-time.
- There is no unified, indigenous “digital twin” for local monitoring, forecasting and scenario analysis.
- Decision-makers (farmers, disaster agencies, planners) lack an accessible, explainable tool.

***In simple words:** All the useful weather/climate data exists, but it is scattered and hard to use together. There is no single, easy screen that fuses it, predicts ahead, and answers “what-if” questions for a region.*

3. Our Solution

Bhū-Darpan is a scalable framework for an AI-driven climate digital twin built on India’s own datasets. It ingests multi-source data, harmonises it onto a common grid, fuses it with data assimilation, generates AI forecasts, and serves everything through a dashboard with a what-if simulator. The pilot focuses on the two key variables the problem statement asks for — **rainfall** and **temperature** — over Jammu & Kashmir.

What the twin delivers

- **Monitor** — current fused climate state: rainfall & temperature anomalies, soil moisture, drought class.
- **Predict** — 14-day AI forecast with a validated skill score and uncertainty band.
- **Simulate** — what-if changes to rainfall/temperature and their impact on drought, crop stress and reservoir inflow.
- **Report** — hazard intelligence (drought, heatwave, monsoon onset) and satellite scene analysis with PDF reports.

4. How It Works — Plain-English Walkthrough

A user opens the dashboard and sees the live climate state. They can switch to the Climate Map to see where anomalies are strongest, open the Forecast tab for the 14-day AI outlook, or drag sliders in the What-If simulator to test scenarios. The Satellite module (login-protected) analyses uploaded imagery.

#	Stage	What happens
1	Data comes in	The system pulls rainfall & temperature grids from national/open feeds and puts them on one common grid.
2	Data is fused	Independent sources are blended with ‘data assimilation’ so errors cancel and confidence increases.
3	AI predicts	A trained deep-learning model (LSTM) forecasts the next 14 days; a safety check keeps it honest.
4	You explore	Maps, charts, what-if sliders and hazard alerts — all in the browser, even offline.

5. Feature Walkthrough

Overview & Live State

Current rainfall/temperature anomalies, soil-moisture index, drought class, warming trend, and a live ‘Live · <date>’ data badge you can click to refresh.

Climate Map

Per-cell anomaly grid over real satellite imagery (Leaflet + Esri), district markers, and a region switch (J&K / all-India).

Forecast

14-day forecast of temperature (LSTM) and rainfall (gradient-boosted ensemble) with skill vs climatology and an uncertainty band.

What-If Simulator

Adjust rainfall % and temperature °C; see modelled drought risk, crop-water stress and reservoir-inflow change from a physical water balance.

Hazards

SPI-based drought, heatwave detection, and monsoon-onset / JJAS progression from the historical record.

Satellite Module

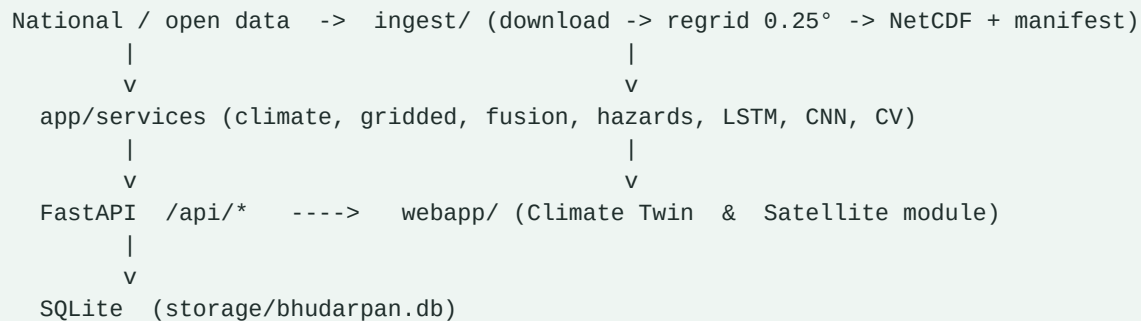
Land-cover classification (EuroSAT CNN), object detection, semantic change detection, and automated PDF reports.

Authentication

Password + email-OTP sign-in; accounts, OTPs and analyses stored in SQLite.

6. Technical Architecture

A single FastAPI backend serves two zero-build single-page apps and a JSON API. Services encapsulate the climate and computer-vision logic; an ETL package handles ingestion; SQLite persists accounts and records.



Technology stack

Layer	Technology
Backend / API	Python 3.12, FastAPI, Uvicorn, Pydantic
Climate AI	PyTorch (LSTM), scikit-learn (gradient-boosted trees), NumPy
Satellite AI / CV	PyTorch CNN (EuroSAT), OpenCV, Pillow
Geo / data	xarray, netCDF4, Leaflet + Esri imagery, Chart.js
Auth / storage	PyJWT, passlib, SMTP OTP, SQLite

7. Data Sources & Pipeline

The ETL (ingest/) downloads a source, decodes it, regrids to a common 0.25° grid, writes NetCDF, and updates a manifest the API reads. Per-cell climatology enables anomaly computation.

Parameter	Source	Status
Gridded rainfall & temperature (live)	NASA POWER (MERRA-2)	Ingested / near-real-time
2nd reanalysis (for fusion)	Open-Meteo (ERA5)	Ingested
Land-surface temperature (LST)	NASA POWER TS (satellite skin temp)	Ingested + fused
INSAT LST / SST / rainfall	ISRO / MOSDAC (3RIMG_L2B_*)	Reader ready; credential-gated
IMD gridded (0.25° / 1.0°)	IMD Pune	Adapter ready; server-gated
Land-cover training set	EuroSAT (Sentinel-2)	Used to train the CNN

Honest note: IMD and the authenticated INSAT download depend on external servers/credentials; the twin currently runs on POWER + ERA5 with satellite skin-temperature fused in, and both gated adapters are wired to drop in.

8. AI / ML Models

8.1 Temperature forecasting — PyTorch LSTM (deep learning)

A 30-day → 14-day sequence model that learns the day-of-year *anomaly* of mean temperature and rainfall. Learning anomalies (not raw values) is what lets it beat a climatology baseline; at inference the

predicted anomaly is added back to each target day’s climatology. Validated on a held-out 3-year split.

8.2 Skill-gating

Model choice is skill-gated: the LSTM drives a variable only when its held-out skill beats climatology. Temperature (predictable) uses the LSTM (+15.7%); rainfall (chaotic at 14 days) falls back to a gradient-boosted ensemble — so a forecast is never worse than climatology.

8.3 Land cover — EuroSAT-trained CNN

A small convolutional network trained on the EuroSAT benchmark (27,000 Sentinel-2 tiles). Held-out test accuracy 95.9% (10-class) / 98.2% grouped into the app’s five classes. Sliding-window inference plus a spectral + texture refinement gives realistic, high-resolution land-cover maps.

8.4 Data assimilation & satellite fusion

Cressman successive-correction and optimal-interpolation fuse independent reanalyses and assimilate the satellite LST field into the air-temperature analysis, quantifying how much the satellite adjusted the twin.

8.5 Change detection

Detects genuine land-cover conversions (water loss, urbanisation) via semantic class-change confirmed by a spectral difference; seasonal colour shifts (crop greening) are excluded so the change % is meaningful.

Model	Task	Validated result
LSTM (PyTorch)	Temperature forecast	+15.7% skill vs climatology (14-day)
GBT ensemble	Rainfall forecast	1-step skill; 14-day fallback
CNN (EuroSAT)	Land-cover classification	95.9% / 98.2% test accuracy
Cressman / OI	Data assimilation	~15% error reduction vs best source

9. Near-Real-Time & Key API Endpoints

A refresh service pulls the latest available reanalysis window on demand (or on a schedule), updates the NetCDF + manifest and clears caches so the Climate Map evolves. Selected endpoints:

Endpoint	Purpose
GET /api/climate/current	Current fused anomalies, soil, drought
GET /api/climate/forecast?var=	14-day AI forecast + skill
POST /api/climate/whatif	Scenario impacts
GET /api/gridded/anomaly?var=®ion=	Per-cell anomaly grid
POST /api/ingest/refresh	Pull latest data (near-real-time)
GET /api/fusion/surface-temp	Assimilate satellite LST into the state
POST /api/analysis & middot; /api/change	Satellite land-cover & middot; change detection
POST /api/auth/register & middot; verify-otp	Account sign-up with email OTP

10. Setup & Run

```
cd Backend
python -m venv .venv
.venv\Scripts\activate      (Windows)
pip install -r requirements.txt
python run.py
```

Open **http://localhost:8000** for the Climate Digital Twin (no login) and **http://localhost:8000/satellite** for the Satellite module (demo login demo@bhudarpan.ai / demo1234). Interactive API docs at **/docs**. Trained model weights and sample grids are bundled, so it works offline; internet is only needed to refresh live data.

11. Alignment with the Problem Statement

Objective / Criterion	How Bhū-Darpan addresses it
Scalable AI digital-twin framework	Modular FastAPI services; two regions (J&K, India)
PoC for rainfall & temperature	Live state + 14-day AI forecast over J&K
Interactive geospatial visualisation	Leaflet map dashboard with per-cell anomaly grids
What-if simulation module	Physical water-balance scenario simulator
AI/ML/DL (TensorFlow/PyTorch)	PyTorch LSTM (climate) + PyTorch CNN (satellite)
Data usage & assimilation	Multi-source fusion + satellite LST assimilation
Validation	Held-out skill scores / test accuracy reported

12. Honest Limitations & Future Work

- IMD gridded ingestion and the authenticated INSAT/MOSDAC download depend on external servers/credentials; adapters are implemented and ready.
- Rainfall remains hard to predict at 14 days — skill-gated to climatology rather than overclaimed.
- Near-real-time is bounded by reanalysis latency (a few days); true streaming and national scale-out are next steps.
- Per-cell deep learning and longer forecast leads are future increments.

Appendix: Project Structure

```
Bhu-Darpan/
  README.md  LICENSE  docs/ARCHITECTURE.md
  Backend/
    run.py  requirements.txt  .env.example
  app/
    main, config, database, security, schemas
  routers/
    auth, analysis, change, climate, gridded, hazards, fusion, ingest, dashb
oard
  services/
    climate, lstm_forecast, ml_forecast, gridded_analysis, fusion,
    satellite_fusion, refresh, hazards, landcover_model, segmentation,
    detection, change_detection, engine, email_service
  ingest/
    ETL (sources/, regrid, pipeline)
  scripts/
    train_landcover.py, train_lstm.py, populate.py
  tests/
    full_test.py, selftest.py
  webapp/
    index.html, satellite.html, assets
```

models/ data/ storage/
frontend/ optional React UI

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